TEXORA · LECTURE SHEET

# Photo Sensors and Transducers

*Devices that turn the physical world into electrical signals: photo sensors — LDR, photodiode and phototransistor — and transducers in general, with the measurement of strain, temperature, pressure, speed and torque worked through device by device.*

EEE-2203 · BASICS OF ELECTRICAL & ELECTRONICS ENGINEERING · TOPIC 6 · LEVEL 2 TERM 2

## 1. Photo sensors

A **photo sensor** converts light into an electrical quantity. All of them exploit the fact that light photons striking a semiconductor release charge carriers (the photoelectric effect inside the material).

| DEVICE | PRINCIPLE | TYPICAL USES |
|---|---|---|
| LDR (light-dependent resistor) | A photoconductive cell, usually cadmium sulphide (CdS): its resistance falls as illumination rises — from about 1 MΩ in darkness to a few hundred ohms in bright light. Slow (tens of milliseconds) and without a junction. | Automatic street lights, light meters, alarm and presence sensing where speed does not matter. |
| Photodiode | A p–n junction operated in reverse bias: light creates extra electron–hole pairs in the depletion region, so the reverse current is proportional to the illumination. Very fast (nanoseconds) and linear. | Optical encoders, barcode readers, remote-control receivers, precision light measurement, fibre-optic links. |
| Phototransistor | A transistor whose base region is exposed to light: the photocurrent acts as base current and is multiplied by β, giving much higher sensitivity than a photodiode, at the cost of speed (microseconds). | Slotted opto-switches, object counters, shaft position and speed sensing. |
| Photovoltaic cell (solar cell) | A large-area junction used unbiased: light generates a voltage and current directly — it is a source, not just a sensor. | Power generation, calculators, and the energy-harvesting supplies of Topic 7. |

A light source (often an LED) paired with a photo sensor forms an **opto-coupler/opto-isolator**: signals cross between circuits with complete electrical isolation — used, for example, to sense a rotating shaft or to protect control electronics from power circuits.

## 2. Transducers: definition and classification

A **transducer** is a device that converts energy from one form to another; a **sensor** is a transducer whose input is a physical quantity (strain, temperature, pressure, speed, torque) and whose output — for measurement purposes — is electrical. Classifications used in practice:

- **Active (self-generating)** transducers produce their own electrical output from the measured energy — thermocouple (voltage from heat), piezoelectric crystal (charge from force), photovoltaic cell. They need no power supply.
- **Passive** transducers change an electrical property — resistance, inductance or capacitance — and need an external supply and a measuring circuit such as a bridge: strain gauge, RTD, thermistor, LDR, LVDT.
- By the electrical principle: **resistive, inductive, capacitive, piezoelectric, photoelectric, thermoelectric**.
- Analogue versus digital output; primary sensing element versus the signal-conditioning electronics (bridge, amplifier) that usually follow it — the op-amps of Topic 5 are the standard conditioners.

## 3. Measurement of strain: the strain gauge

When a wire or foil conductor is stretched, it becomes longer and thinner, so its resistance rises. A **strain gauge** — a fine metal-foil grid bonded to a backing — is glued to the surface of a part so that it shares the part's strain $\varepsilon$ ($= \Delta L/L$). The sensitivity is the **gauge factor**:

$$GF = (\Delta R/R)/\varepsilon, \quad GF \approx 2 \text{ for metal-foil gauges}$$

Because $\Delta R$ is tiny, the gauge is wired into a **Wheatstone bridge** (Topic 1): one arm is the active gauge, and the bridge output — a few millivolts — is amplified. Using two or four gauges (a half or full bridge) multiplies sensitivity and cancels temperature effects.

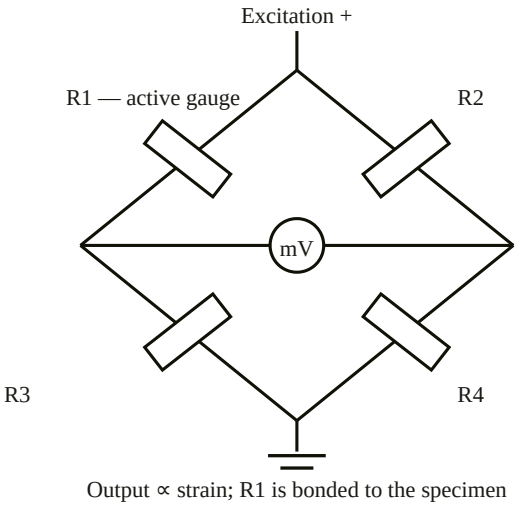



Fig. 1 — Strain gauge in a Wheatstone bridge (quarter-bridge form). The bridge unbalance is proportional to the strain.

**WORKED EXAMPLE — STRAIN GAUGE**

A foil gauge has R = 120 Ω and GF = 2.0. Bonded to a beam strained by 500 microstrain ($\varepsilon = 500 \times 10^{-6}$): $\Delta R/R = GF \times \varepsilon = 2.0 \times 500 \times 10^{-6} = 1.0 \times 10^{-3}$, so $\Delta R = 120 \times 10^{-3} =$ **0.12 Ω** — far too small for an ohmmeter, which is exactly why the bridge is used.

## 4. Measurement of temperature

| DEVICE | PRINCIPLE | RANGE AND CHARACTER |
|---|---|---|
| Thermocouple | Two dissimilar metals joined at a measuring junction; a temperature difference between this junction and the reference junction generates a small EMF (the **Seebeck effect**). Type K (chromel–alumel) gives about 41 μV/°C. Active device. | Very wide range (type K: about −200 to +1250 °C), rugged and cheap; needs reference-junction compensation; millivolt output needs amplification. |
| RTD (resistance temperature detector) | The resistance of a pure metal (usually platinum) rises almost linearly with temperature: $R = R_0(1 + \alpha t)$, $\alpha = 0.00385$/°C for platinum. The Pt100 has $R_0 = 100$ Ω at 0 °C. Passive; measured with a bridge. | About −200 to +650 °C; the most accurate and stable of the three; more expensive. |
| Thermistor | A semiconductor bead whose resistance falls steeply as temperature rises (NTC type) — a large change over a small range, strongly non-linear. | About −50 to +150 °C; very sensitive and cheap; used in thermostats, dryers and protection circuits. |

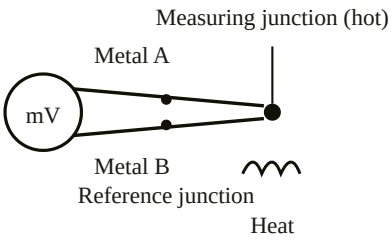



Fig. 2 — Thermocouple: the EMF depends on the difference between the measuring and reference junction temperatures.

**WORKED EXAMPLES — TEMPERATURE SENSORS**

Type K thermocouple, sensitivity 41 µV/°C, measuring junction 200 °C hotter than the reference: EMF = 41 × 200 = 8200 µV = **8.2 mV**.

Pt100 at 100 °C: R = 100(1 + 0.00385 × 100) = 100 × 1.385 = **138.5 Ω**.

## 5. Measurement of pressure

Pressure is usually measured by letting it deflect an elastic element and converting the deflection or the resulting strain into a signal:

- **Bourdon tube:** a flattened, C-shaped tube that straightens as pressure rises; its free end moves a pointer (the classic dial gauge) or a potentiometer/LVDT for an electrical output.
- **Diaphragm with strain gauges:** pressure strains a metal or silicon diaphragm carrying foil or piezoresistive gauges in a bridge — the standard modern pressure transmitter.
- **Piezoelectric sensor:** a crystal that generates charge when stressed; excellent for rapidly changing (dynamic) pressure, unsuitable for steady pressure because the charge leaks away.
- **Capacitive sensor:** pressure moves a diaphragm that forms one plate of a capacitor; the capacitance change is measured electronically.

## 6. Measurement of speed and torque

**Speed.** (1) The **D.C. tachogenerator** is a small generator on the shaft: its output voltage is proportional to speed — a direct analogue reading. An A.C. tachogenerator works similarly with an alternating output. (2) The **optical encoder** passes a slotted disc between an LED and a photo sensor (Section 1): the pulse frequency is proportional to speed, and counting pulses also gives position. (3) A **magnetic pick-up** near a toothed wheel produces a pulse per tooth — the rugged industrial standard on motors and looms.

**Torque.** The standard method bonds strain gauges at 45° to a rotating shaft, arranged as a full bridge so that only the twisting strain is measured; the signal leaves the shaft through slip rings or a rotating

transformer/telemetry link. The angular twist of a shaft under torque is proportional to the torque, so calibrating the bridge calibrates the torque meter. Reaction torque — measuring the force with which a motor's stator tries to rotate against its mounting (force × lever arm) — is the simpler laboratory alternative.

## 7. Key terms

- **Photo sensor:** a device converting light into an electrical signal — photoconductive (LDR), photovoltaic (solar cell) or junction-based (photodiode, phototransistor).
- **Transducer:** a device converting energy between forms; a sensor is a transducer that turns a physical quantity into an electrical signal.
- **Active / passive transducer:** self-generating (thermocouple) versus property-changing and externally powered (strain gauge, RTD).
- **Gauge factor:** the ratio $(\Delta R/R)/\varepsilon$ of a strain gauge, about 2 for metal foil.
- **Seebeck effect:** the EMF generated at a junction of two dissimilar metals when the junction is at a different temperature from the rest of the circuit.
- **Signal conditioning:** the bridge, amplification and filtering that turn a transducer's raw output into a usable measurement.

## 8. Key points

- LDR: resistance falls with light, slow and simple. Photodiode: reverse current proportional to light, fast and linear. Phototransistor: $\beta$ times more sensitive, slower.
- Strain is measured by resistance change — GF ≈ 2 turns microstrain into a tiny $\Delta R$, so a Wheatstone bridge and amplifier are essential.
- Temperature: thermocouple (active, wide range, millivolts), RTD (accurate, linear, Pt100 = 138.5 Ω at 100 °C), thermistor (sensitive, narrow range, non-linear).
- Pressure is converted to deflection or strain first (Bourdon tube, diaphragm); dynamic pressure suits the piezoelectric crystal.
- Speed comes from a tachogenerator's voltage or a pulse train's frequency (optical or magnetic); torque from strain gauges at 45° on the shaft.

## 9. Summary

Measurement begins where a physical quantity is converted into an electrical signal. Light is sensed by the LDR, photodiode and phototransistor; strain by the foil gauge in a bridge; temperature by the thermocouple, the platinum RTD and the thermistor; pressure by elastic elements strained or deflected;

speed by generators and pulse counters; torque by gauges on the twisting shaft. Choosing among them is a matter of range, sensitivity, speed and cost — and almost every chain ends in the bridge and amplifier circuits of the earlier topics. The next topic shows these same devices shrunk into clothing itself.

## 10. Practice

Answer in your own words or by calculation; brief answers follow. No multiple-choice questions.

1. A strain gauge of 350 Ω with GF = 2.1 is strained by 800 microstrain. Find ΔR.
2. Find the resistance of a Pt100 RTD at 50 °C.
3. A type K thermocouple has its measuring junction 150 °C above the reference junction. Estimate the output EMF.
4. Choose a photo sensor for (a) switching a street light at dusk, (b) counting bottles on a fast conveyor, and justify each choice in one line.
5. Classify the thermocouple and the strain gauge as active or passive, with one reason each.

### Answer key

1. $\Delta R/R = GF \times \varepsilon = 2.1 \times 800 \times 10^{-6} = 1.68 \times 10^{-3}$. $\Delta R = 350 \times 1.68 \times 10^{-3} = 0.588\ \Omega$.
2. $R = 100(1 + 0.00385 \times 50) = 100 \times 1.1925 = 119.25\ \Omega \approx 119.3\ \Omega$.
3. EMF ≈ 41 μV/°C × 150 °C = 6150 μV = 6.15 mV.
4. (a) LDR — cheap, slow response is acceptable, and it works in a simple divider circuit. (b) Photodiode (or phototransistor in a slotted opto-switch) — its fast response can follow objects passing at conveyor speed, which an LDR cannot.
5. Thermocouple: active — it generates its own EMF from the heat energy at the junction. Strain gauge: passive — it only changes resistance and needs an external bridge supply to produce a signal.